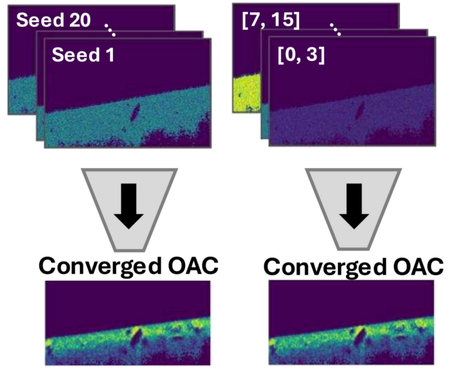
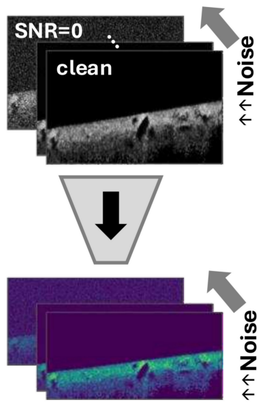
(a) Seed Sweep **(b) In. Range Sweep**



(c) Identifiability Summary

In-vitro Phantom Acq.	Concentration and FP	Seed CoV (20 seeds) [%]	Init CoV (selected) [%]
A0	1.0% (F1)	0.23	13.1
A1	1.0% (F2)	0.18	8.3
A2	1.5% (F1)	0.31	7.6
A3	1.5% (F2)	0.31	5.9
A4	2.0% (F1)	0.29	7.8
A5	2.0% (F2)	0.28	4.7
A6	3.0% (F1)	0.32	6.6
A7	3.0% (F2)	0.27	4.7
A8	5.0% (F1)	0.24	2.8
A9	5.0% (F2)	0.28	2.1

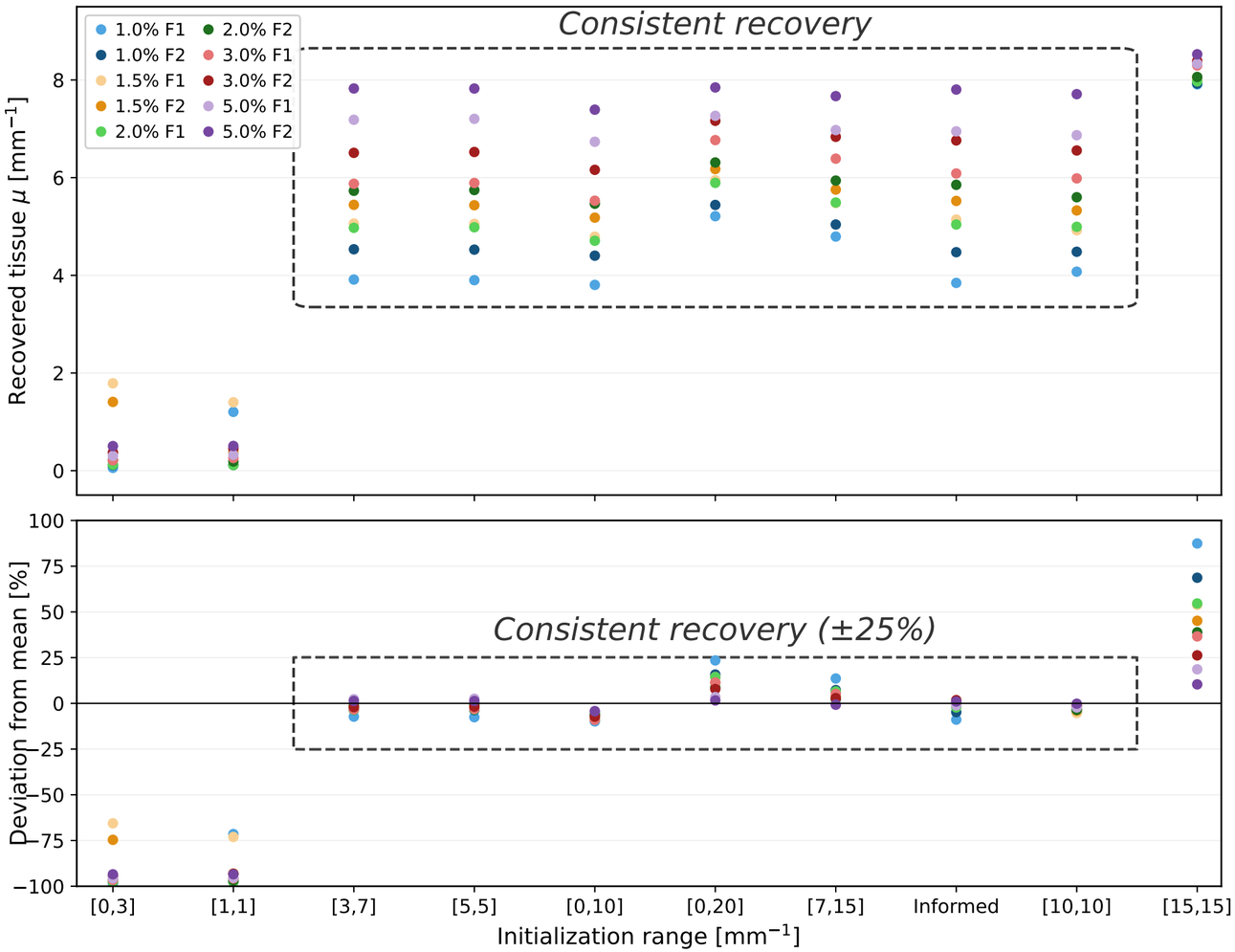
(e) SNR Sweep



(f) Robustness Summary

SNR [dB]	$\lvert\Delta\mu\rvert$ vs baseline (%)	R^2 (μ vs SNR curve)	Max σ_μ [mm^{-1}]
Original	0 (ref)	1.000	–
16	8.5 ± 8.2	0.915	1.385
12	19.5 ± 12.9	0.778	1.922
8	34.6 ± 11.2	0.585	2.190
4	51.8 ± 6.9	0.362	2.336
0	63.8 ± 11.5	0.197	2.554

(d) Recovered OAC across initialization ranges



(g) Per-acquisition Noise Degradation

